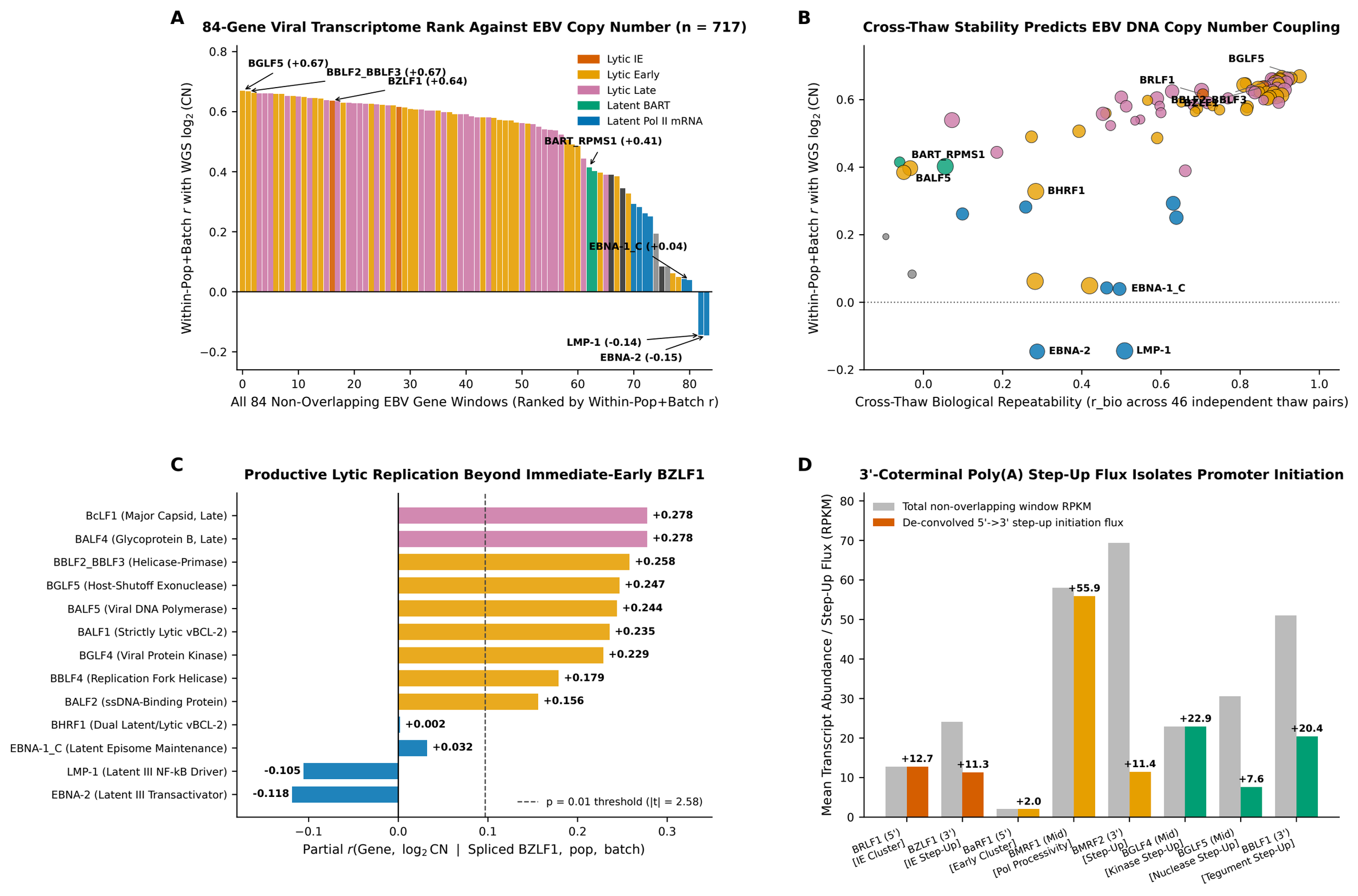

A
84-Gene Viral Transcriptome Rank Against EBV Copy Number (n = 717)
Within-Pop+Batch r with WGS log2 (CN)
All 84 Non-Overlapping EBV Gene Windows (Ranked by Within-Pop+Batch r)
Lytic IE
Lytic Early
Lytic Late
Latent BART
Latent Pol II mRNA
BGLF5 (+0.67)
BBLF2_BBLF3 (+0.67)
BZLF1 (+0.64)
BART_RPMS1 (+0.41)
EBNA-1_C (+0.04)
LMP-1 (-0.14)
EBNA-2 (-0.15)
B
Cross-Thaw Stability Predicts EBV DNA Copy Number Coupling
Within-Pop+Batch r with WGS log2 (CN)
Cross-Thaw Biological Repeatability (r_bio across 46 independent thaw pairs)
BGLF5
BRLF1
BBLF2_BBLF3
BART_RPMS1
BALF5
BHRF1
EBNA-1_C
EBNA-2
LMP-1
C
Productive Lytic Replication Beyond Immediate-Early BZLF1
BcLF1 (Major Capsid, Late)
BALF4 (Glycoprotein B, Late)
BBLF2_BBLF3 (Helicase-Primase)
BGLF5 (Host-Shutoff Exonuclease)
BALF5 (Viral DNA Polymerase)
BALF1 (Strictly Lytic vBCL-2)
BGLF4 (Viral Protein Kinase)
BBLF4 (Replication Fork Helicase)
BALF2 (ssDNA-Binding Protein)
BHRF1 (Dual Latent/Lytic vBCL-2)
EBNA-1_C (Latent Episome Maintenance)
LMP-1 (Latent III NF-kB Driver)
EBNA-2 (Latent III Transactivator)
+0.278
+0.278
+0.258
+0.247
+0.244
+0.235
+0.229
+0.179
+0.156
+0.002
+0.032
-0.105
-0.118
p = 0.01 threshold (|t| = 2.58)
Partial r(Gene, log2 CN | Spliced BZLF1, pop, batch)
D
3'-Coterminal Poly(A) Step-Up Flux Isolates Promoter Initiation
Mean Transcript Abundance / Step-Up Flux (RPKM)
Total non-overlapping window RPKM
De-convolved 5'->3' step-up initiation flux
+12.7
+11.3
+2.0
+55.9
+11.4
+22.9
+7.6
+20.4
BRLF1 (5') [IE Cluster]
BZLF1 (3') [IE Step-Up]
BaRF1 (5') [Early Cluster]
BMRF1 (Mid) [Pol Processivity]
BMRF2 (3') [Step-Up]
BGLF4 (Mid) [Kinase Step-Up]
BGLF5 (Mid) [Nuclease Step-Up]
BBLF1 (3') [Tegument Step-Up]